

Δ vs frozen (good-oriented, value + 95% CI) · PRETRAIN block / SURGERY block · SCORE = mean per-metric-normalized (0-1) · BLUE BOX + big Δ = best arm per metric
WINNER row (champion duel — S=surgery, P=pretrain, ==tie): surgery 5 · pretrain 2 · tie 2 → SURGERY

ViT-g pretrain_2X encoder	+5.81 [+3.67, +8.05]	+0.04 [+0.04, +0.04]	-0.03 [-0.04, -0.01]	0 [0, 0]	0 [0, 0]	0 [0, 0]	+0.01 [+0.01, +0.01]	0 [0, 0]	0.49	
ViT-g pretrain encoder	+4.00 [+1.70, +6.36]	+0.04 [+0.03, +0.04]	-0.02 [-0.04, -0.01]	0 [0, +0.01]	0 [0, 0]	0 [0, 0]	+0.01 [+0.01, +0.01]	0 [0, 0]	0.60	
ViT-g pretrain head	+0.33 [-1.75, +2.41]	0 [0, 0]	-0.01 [-0.02, 0]	0 [0, 0]	0 [+nan, +nan]	0 [+nan, +nan]	0 [+nan, +nan]	0 [+nan, +nan]	0.30	
ViT-g surgery 3stage_DI encoder	+5.37 [+3.23, +7.56]	+0.04 [+0.04, +0.04]	-0.03 [-0.04, -0.01]	+0.01 [+0.01, +0.01]	0 [0, 0]	0 [0, +0.01]	0 [0, 0]	0 [0, 0]	0.54	
ViT-g surgery 3stage_DI head	+4.05 [+1.92, +6.25]	+0.04 [+0.04, +0.05]	-0.02 [-0.03, -0.01]	0 [0, +0.01]	0 [0, 0]	0 [0, 0]	0 [0, 0]	+0.01 [+0.01, +0.01]	0.62	
ViT-g surgery noDI encoder	+4.22 [+1.97, +6.63]	+0.05 [+0.05, +0.05]	-0.02 [-0.04, -0.01]	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]	+0.01 [+0.01, +0.01]	0.59	
ViT-g surgery noDI head	+4.71 [+2.36, +7.07]	+0.06 [+0.05, +0.06]	-0.02 [-0.04, -0.01]	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]	+0.01 [+0.01, +0.01]	0.65	
WINNER	P	S	P	S	=	S	=	S		
	action top1 ↑ better classify motion class from frozen features	motion_cos ↑ better same-motion clips cluster vs different motions	taxonomy_f1 ↑ better read scene attributes from features	future_rmse ↓ better predictor guesses masked next-frame tokens	rollout ↓ better error growth over multi-step rollout	causal ↓ better predict 2nd half from 1st half	tdist ↓ better accuracy decay vs prediction horizon	teacher_free ↓ better free-run minus teacher-forced gap	maskratio ↓ better degradation under heavier masking	SCORE (norm 0-1)